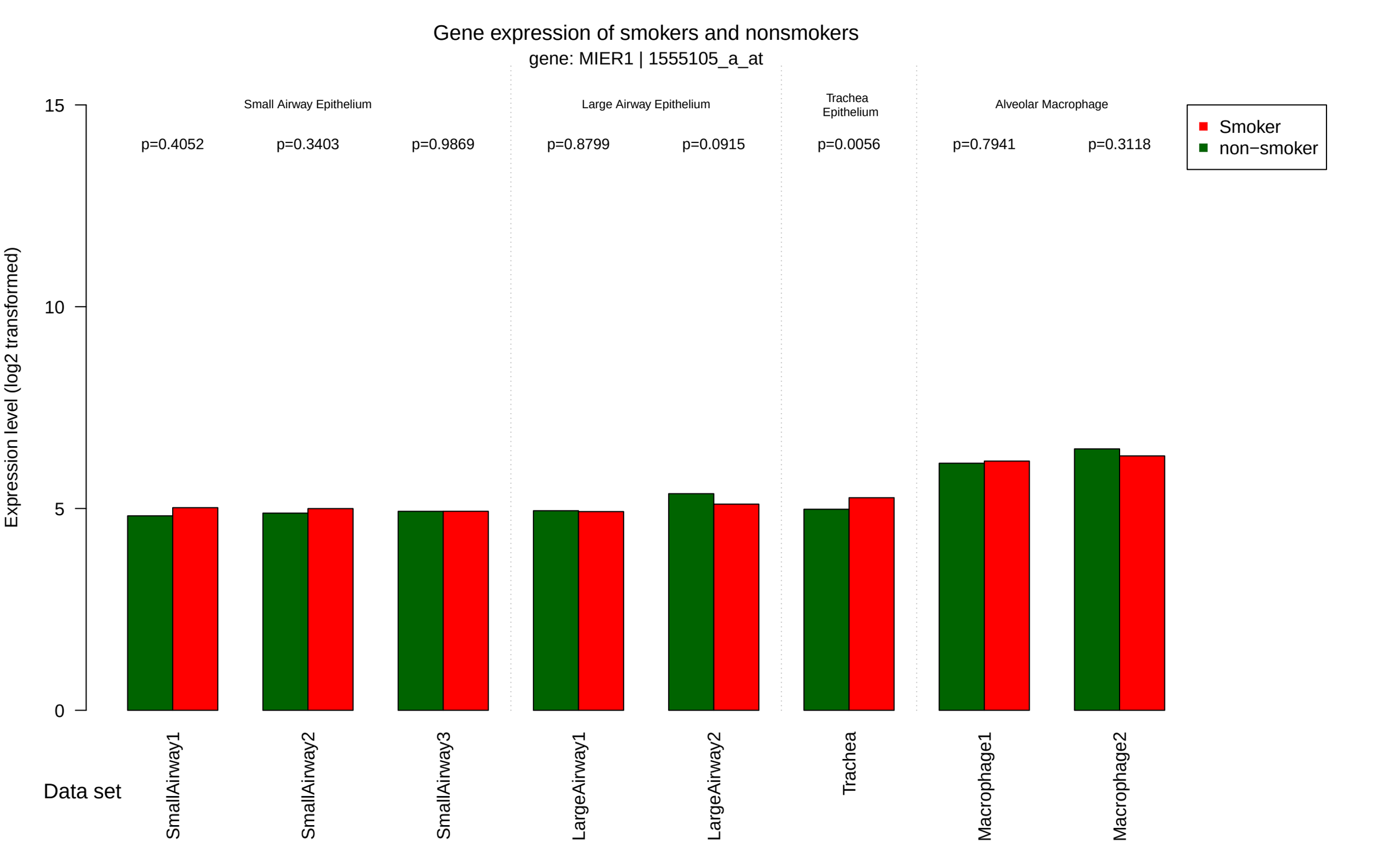
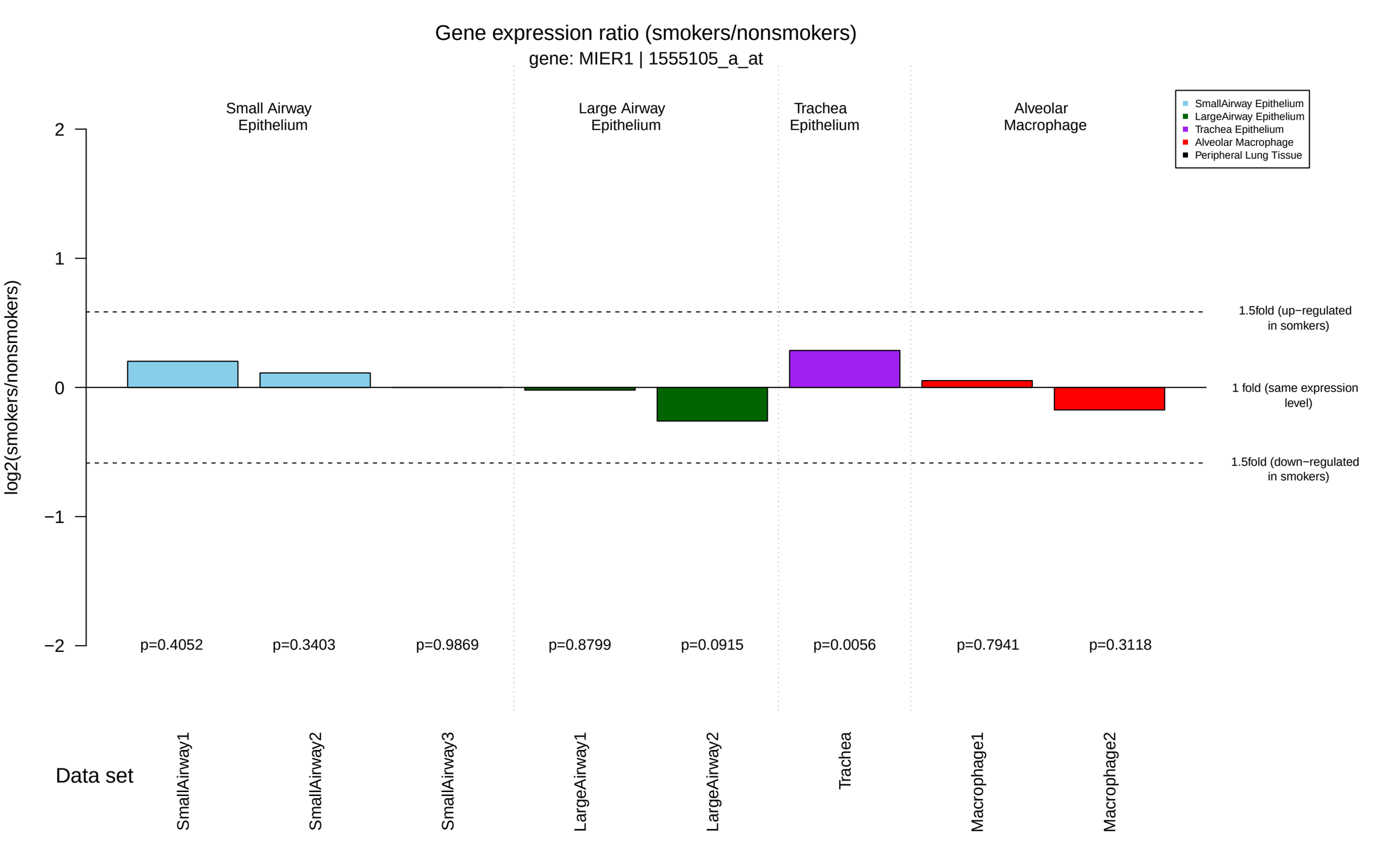
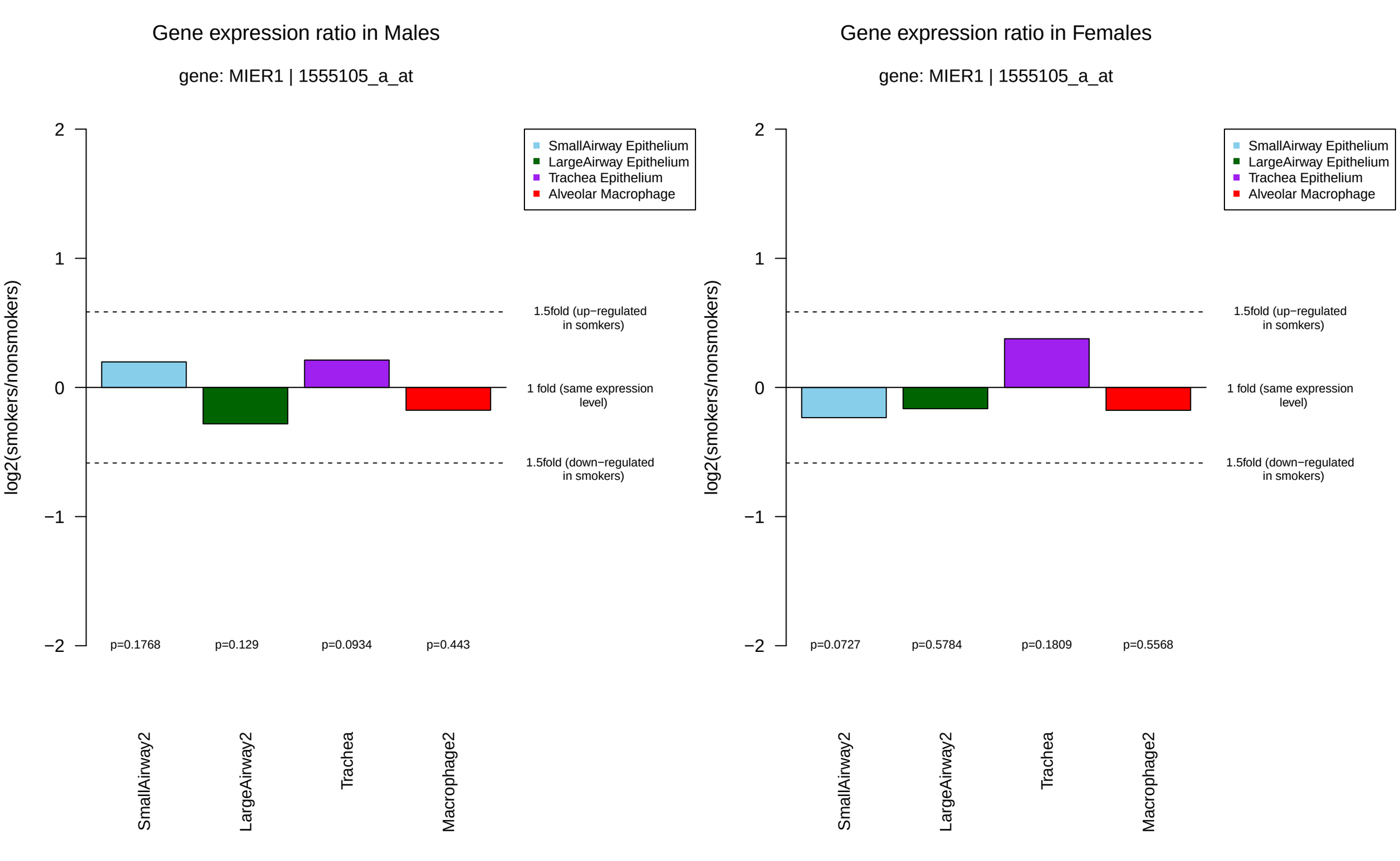
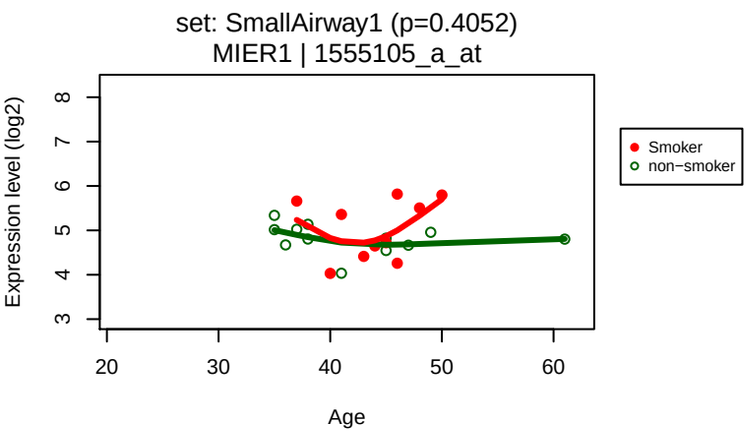
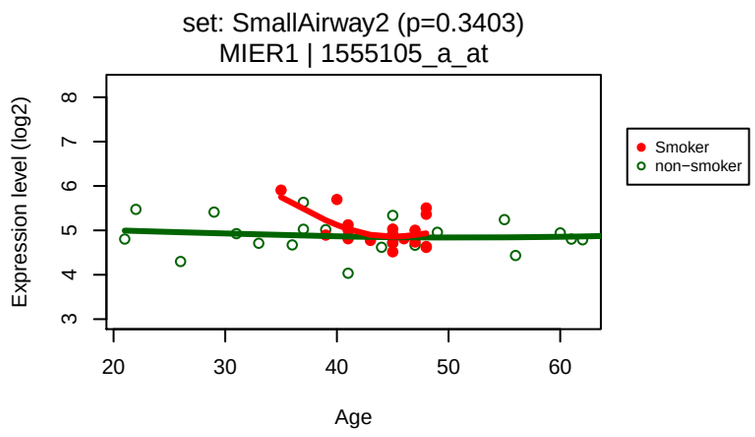
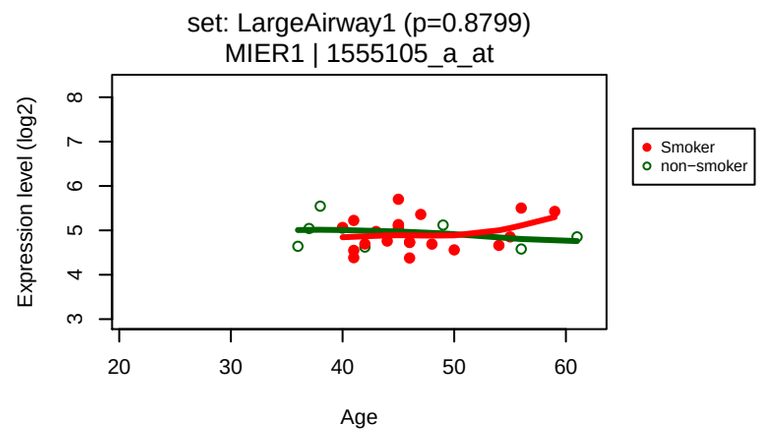

Gene expression of smokers and nonsmokers
gene: MIER1 | 1555105_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4052
p=0.3403
p=0.9869
p=0.8799
p=0.0915
p=0.0056
p=0.7941
p=0.3118
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

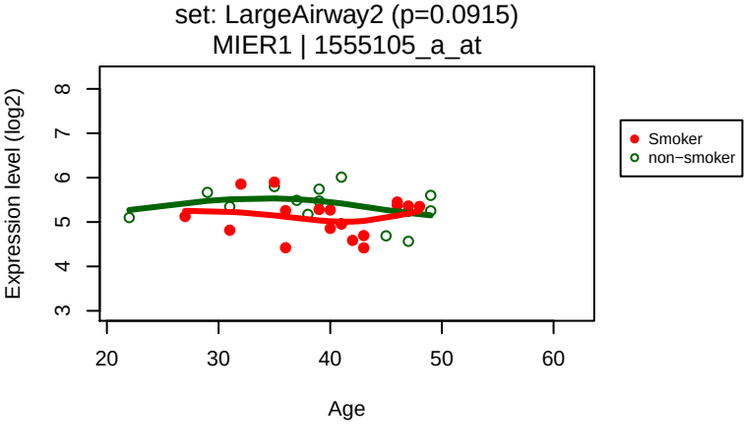

Gene expression ratio (smokers/nonsmokers)
gene: MIER1 | 1555105_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4052
p=0.3403
p=0.9869
p=0.8799
p=0.0915
p=0.0056
p=0.7941
p=0.3118
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

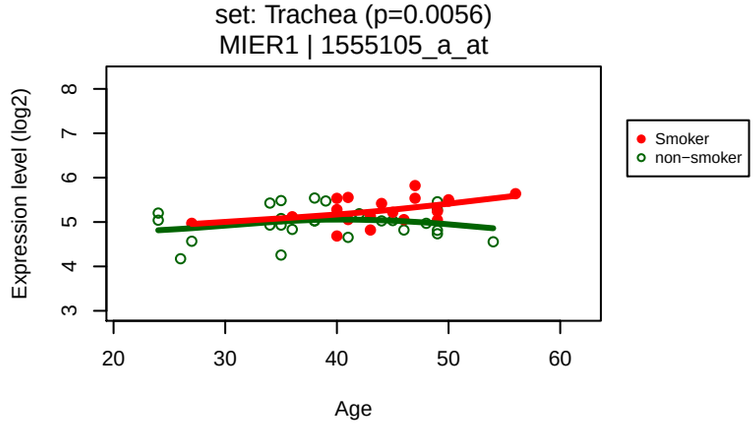

Gene expression ratio in Males
gene: MIER1 | 1555105_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1768
p=0.129
p=0.0934
p=0.443
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MIER1 | 1555105_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0727
p=0.5784
p=0.1809
p=0.5568
SmallAirway2
LargeAirway2
Trachea
Macrophage2

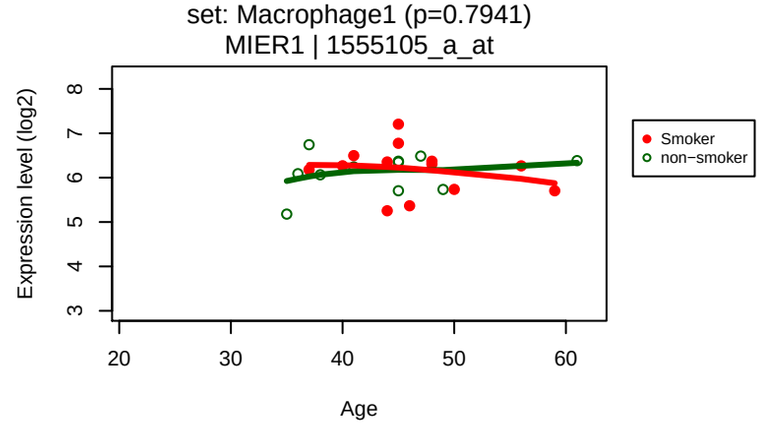

set: SmallAirway1 (p=0.4052)
MIER1 | 1555105_a_at
Expression level (log2)
Age
Smoker
non−smoker

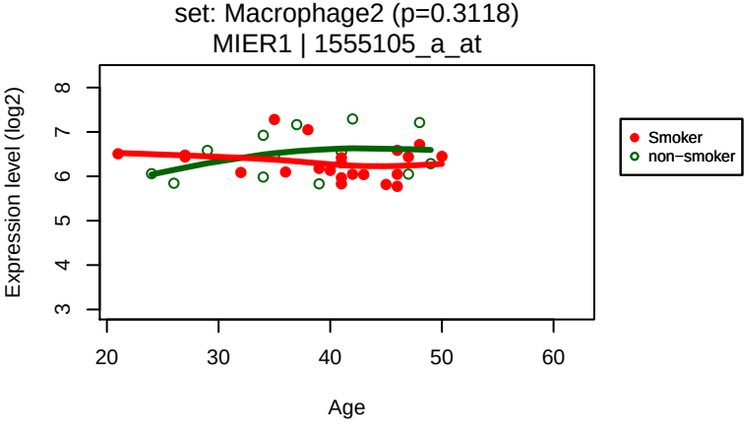

set: SmallAirway2 (p=0.3403)
MIER1 | 1555105_a_at
Expression level (log2)
Age
Smoker
non−smoker

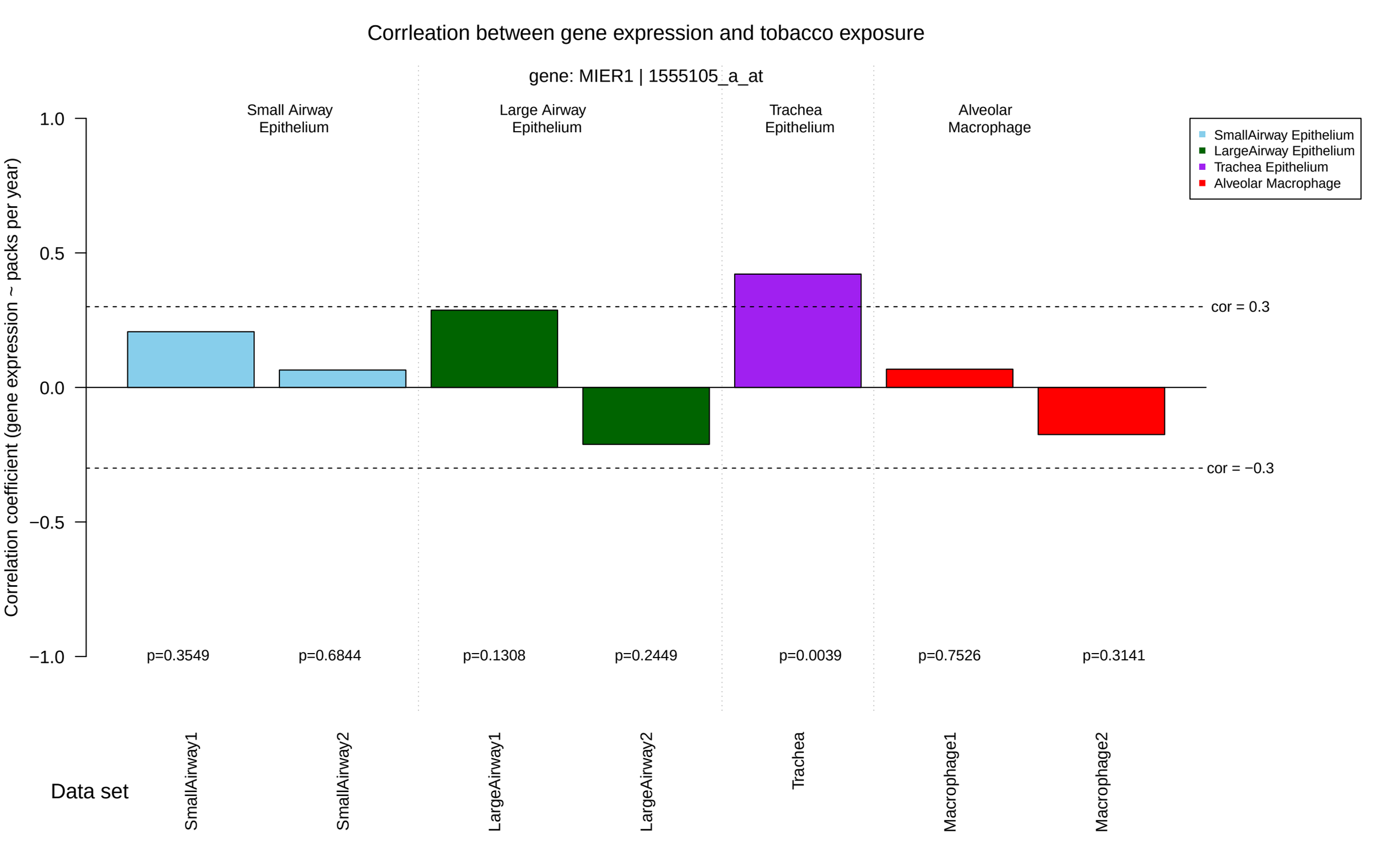

set: LargeAirway1 (p=0.8799)
MIER1 | 1555105_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0915)
MIER1 | 1555105_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0056)
MIER1 | 1555105_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7941)
MIER1 | 1555105_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3118)
MIER1 | 1555105_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MIER1 | 1555105_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3549
p=0.6844
p=0.1308
p=0.2449
p=0.0039
p=0.7526
p=0.3141
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2